Problem 6

Problem 6

Problem 1 *Evaluate the following limits.*

(a) $\lim\limits_{x\to\infty} \dfrac{\sqrt[3]{x^9+5}}{3x^3+\sqrt{4x^6+1}}$

Explanation. This limit is of the form: $\dfrac{\infty}{\infty}$.

This will be a very detailed solution. Make sure you can justify every step below.

$$
\begin{aligned}
\lim_{x\to\infty} \frac{\sqrt[3]{x^9+5}}{3x^3+\sqrt{4x^6+1}} &= \lim_{x\to\infty} \frac{\sqrt[3]{x^9+5}}{3x^3+\sqrt{4x^6+1}} \cdot \frac{\frac{1}{x^3}}{\frac{1}{x^3}} \\
&= \lim_{x\to\infty} \frac{\frac{\sqrt[3]{x^9+5}}{x^3}}{\frac{3x^3+\sqrt{4x^6+1}}{x^3}} \\
&= \lim_{x\to\infty} \frac{\sqrt[3]{\frac{x^9+5}{x^9}}}{\frac{3x^3}{x^3}+\sqrt{\frac{4x^6+1}{x^6}}} \\
&= \lim_{x\to\infty} \frac{\sqrt[3]{1+\frac{5}{x^9}}}{3+\sqrt{4+\frac{1}{x^6}}} \\
&= \frac{\lim\limits_{x\to\infty} \sqrt[3]{1+\dfrac{5}{x^9}}}{\lim\limits_{x\to\infty}\left(3+\sqrt{4+\dfrac{1}{x^6}}\right)} \\
&= \frac{\sqrt[3]{\lim\limits_{x\to\infty}\left(1+\dfrac{5}{x^9}\right)}}{3+\lim\limits_{x\to\infty}\sqrt{4+\dfrac{1}{x^6}}} \\
&= \frac{\sqrt[3]{1+\lim\limits_{x\to\infty}\dfrac{5}{x^9}}}{3+\sqrt{\lim\limits_{x\to\infty}\left(4+\dfrac{1}{x^6}\right)}} \\
&= \frac{\sqrt[3]{1+5\lim\limits_{x\to\infty}\left(\dfrac{1}{x}\right)^9}}{3+\sqrt{4+\lim\limits_{x\to\infty}\dfrac{1}{x^6}}} \\
&= \frac{\sqrt[3]{1+5\left(\lim\limits_{x\to\infty}\dfrac{1}{x}\right)^9}}{3+\sqrt{4+\left(\lim\limits_{x\to\infty}\dfrac{1}{x}\right)^6}} \\
&= \frac{\sqrt[3]{1+5\left(0\right)^9}}{3+\sqrt{4+\left(0\right)^6}} \\
&= \frac{1}{3+\sqrt{4}} \\
&= \frac{1}{5}
\end{aligned}
$$

(b) $\lim\limits_{x\to\infty} \dfrac{\sin(9x)}{5x}$

Explanation. Since $-1 \leq \sin(9x) \leq 1$ for all x , and we can divide by $5x > 0$, $\dfrac{-1}{5x} \leq \dfrac{\sin(9x)}{5x} \leq \dfrac{1}{5x}$.
We have $\lim\limits_{x\to\infty} \left(-\dfrac{1}{5x}\right) = -\dfrac{1}{5} \lim\limits_{x\to\infty} \left(\dfrac{1}{x}\right) = 0$ and $\lim\limits_{x\to\infty} \dfrac{1}{5x} = 0$. The Squeeze theorem implies that

$$\lim_{x\to\infty} \frac{\sin(9x)}{5x} = 0$$